



**The University of Azad Jammu and Kashmir,
Muzaffarabad**

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Roll No: 2024-SE-06

Course Title: Database System

Course Code: SE-2204

Lab No : 06

Instructor Name: Awais Rathor

Department Of Software Engineering

✓ **Filters**

1. Introduction

This lab was performed to understand SQL filtering techniques using MySQL in XAMPP. Different SQL operators and clauses such as WHERE, AND, OR, BETWEEN, IN, LIKE, IS NULL, ORDER BY, and LIMIT were used to retrieve specific employee records from the Employee table.

2. Objective

The objective of this lab was to practice SQL filtering and sorting operations on a database table using MySQL queries.

3. Database Creation

First, the XAMPP server was started and phpMyAdmin was opened through localhost. A database named “**companydb**” was created successfully.



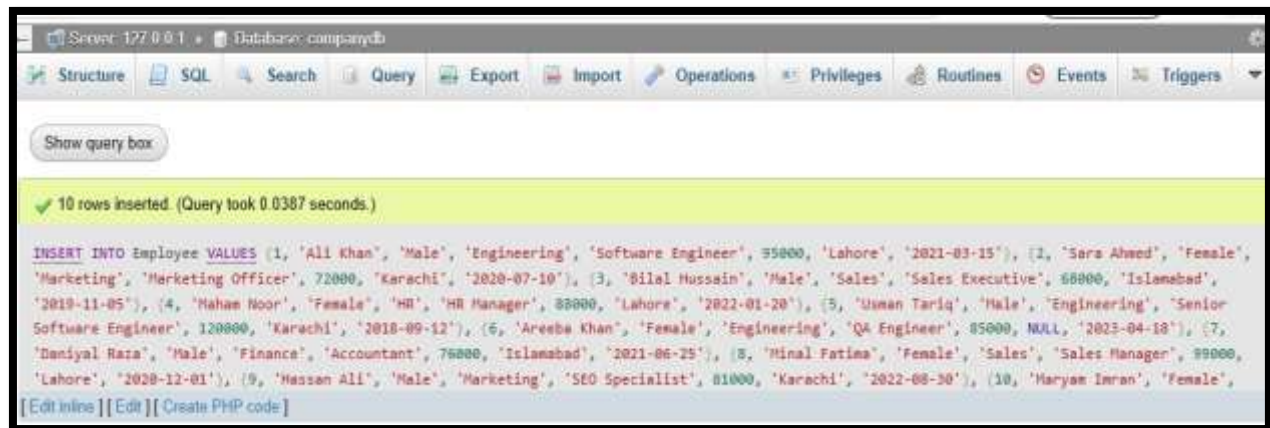
4. Employee Table Creation

An Employee table was created in the company-db database using **SQL CREATE TABLE** statement.



Filters Comparison & Logical Operators

Sample employee records were inserted into the Employee table using INSERT INTO statement.



```
Server: 127.0.0.1 • Database: companydb
Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers
Show query box
10 rows inserted (Query took 0.0387 seconds)
INSERT INTO Employee VALUES (1, 'Ali Khan', 'Male', 'Engineering', 'Software Engineer', 95000, 'Lahore', '2021-03-15'), (2, 'Sara Ahmed', 'Female', 'Marketing', 'Marketing Officer', 72000, 'Karachi', '2020-07-10'), (3, 'Bilal Hussain', 'Male', 'Sales', 'Sales Executive', 66000, 'Islamabad', '2019-11-05'), (4, 'Maham Noor', 'Female', 'HR', 'HR Manager', 88000, 'Lahore', '2022-01-20'), (5, 'Usman Tariq', 'Male', 'Engineering', 'Senior Software Engineer', 120000, 'Karachi', '2018-09-12'), (6, 'Areeba Khan', 'Female', 'Engineering', 'QA Engineer', 85000, NULL, '2023-04-18'), (7, 'Daniyal Raza', 'Male', 'Finance', 'Accountant', 76000, 'Islamabad', '2021-06-25'), (8, 'Minal Fatima', 'Female', 'Sales', 'Sales Manager', 99000, 'Lahore', '2020-12-01'), (9, 'Hassan Ali', 'Male', 'Marketing', 'SEO Specialist', 81000, 'Karachi', '2022-08-30'), (10, 'Maryam Imran', 'Female', 'Marketing', 'Marketing Officer', 72000, 'Karachi', '2020-07-10')
[Edit inline] [Edit] [Create PHP code]
```

5. Part A Queries

➤ Task A1

- This query displays employee ID, employee name, and salary of employees earning more than 90,000.

SELECT EmpID, EmpName, Salary

FROM Employee

WHERE Salary > 90000;



```
-- Task A1 SELECT EmpID, EmpName, Salary FROM Employee WHERE Salary > 90000;
[Edit inline] [Edit] [Create PHP code]
Show all | Number of rows: 25 | Filter rows: Search this table
Extra options
Showing rows 0 - 3 (4 total, Query took 0.0005 seconds)
-- Task A1 SELECT EmpID, EmpName, Salary FROM Employee WHERE Salary > 90000;
[Edit inline] [Edit] [Create PHP code]
Show all | Number of rows: 25 | Filter rows: Search this table
Extra options
```

	EmpID	EmpName	Salary
☐ Edit ☐ Copy ☐ Delete	1	Ali Khan	95000.00
☐ Edit ☐ Copy ☐ Delete	5	Usman Tariq	120000.00
☐ Edit ☐ Copy ☐ Delete	8	Minal Fatima	99000.00
☐ Edit ☐ Copy ☐ Delete	10	Maryam Imran	105000.00

↑ ☐ Check all With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

➤ Task A2

- This query shows employees whose salary is less than or equal to 75,000.

SELECT EmpName, Salary

FROM Employee

WHERE Salary <= 75000;

Showing rows 0 - 1 (2 total, Query took 0.0015 seconds.)

```
-- Task A2 SELECT EmpName, Salary FROM Employee WHERE Salary <= 75000;
```

Edit inline][Edit][Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

				EmpName	Salary
☐	Edit	Copy	Delete	Sara Ahmed	72000.00
☐	Edit	Copy	Delete	Bilal Hussain	68000.00

↑ ☐ Check all With selected: Edit Copy Delete Export

Task A2 query showing employees with salary less than or equal to 75,000.

➤ Task A3

- This query finds employees working in Lahore with salary greater than 90,000.

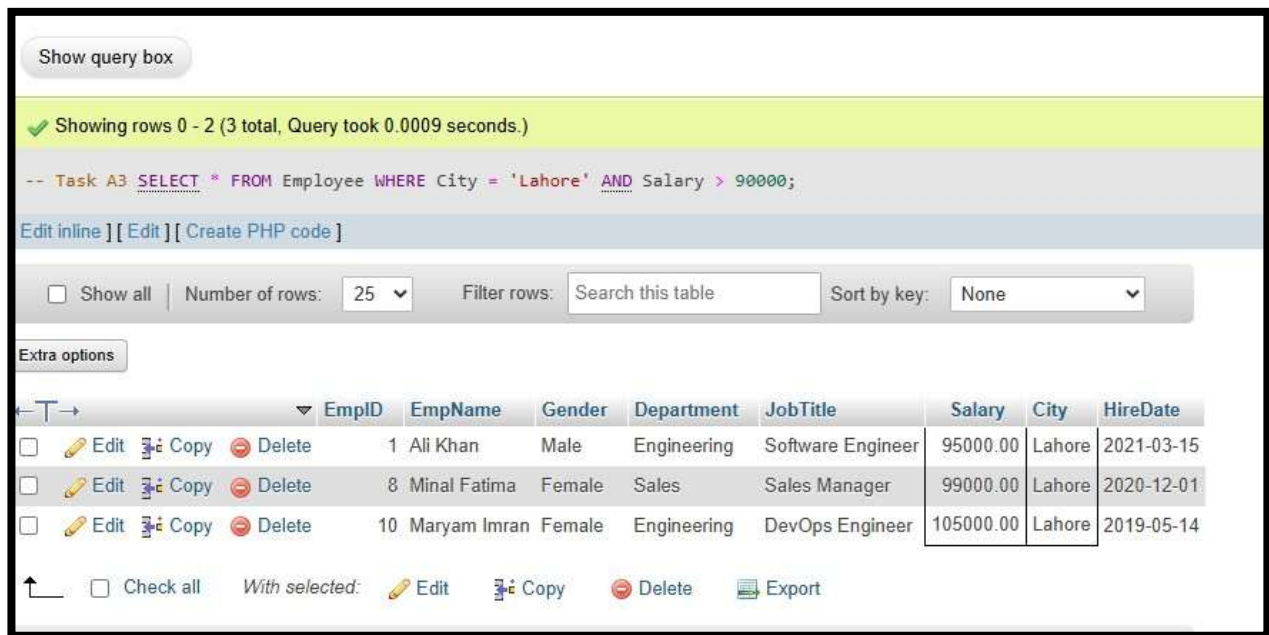
SELECT *

FROM Employee

Filters Comparison & Logical Operators

WHERE City = 'Lahore'

AND Salary > 90000;



The screenshot shows a database query interface. At the top, there is a button labeled "Show query box". Below it, a green status bar indicates "Showing rows 0 - 2 (3 total, Query took 0.0009 seconds.)". The query text is displayed as: `-- Task A3 SELECT * FROM Employee WHERE City = 'Lahore' AND Salary > 90000;`. Below the query, there are links for "Edit inline", "[Edit]", and "[Create PHP code]". A control bar includes a "Show all" checkbox, a "Number of rows" dropdown set to 25, a "Filter rows" search box with the placeholder "Search this table", and a "Sort by key" dropdown set to "None". An "Extra options" button is also present. The main area displays a table with 9 columns: EmpID, EmpName, Gender, Department, JobTitle, Salary, City, and HireDate. Three rows are visible, each with edit, copy, and delete icons. At the bottom, there is a "Check all" checkbox and a "With selected:" section containing edit, copy, delete, and export icons.

EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate
1	Ali Khan	Male	Engineering	Software Engineer	95000.00	Lahore	2021-03-15
8	Minal Fatima	Female	Sales	Sales Manager	99000.00	Lahore	2020-12-01
10	Maryam Imran	Female	Engineering	DevOps Engineer	105000.00	Lahore	2019-05-14

Task A3 query showing employees from Lahore earning above 90,000.

➤ Task A4

- This query lists employees belonging to Karachi or Islamabad.

SELECT EmpName, City

FROM Employee

WHERE City = 'Karachi'

OR City = 'Islamabad';

Filters Comparison & Logical Operators

✓ Showing rows 0 - 4 (5 total, Query took 0.0005 seconds.)

```
-- Task A4 SELECT EmpName, City FROM Employee WHERE City = 'Karachi' OR City = 'Islamabad';
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

				EmpName	City
☐	Edit	Copy	Delete	Sara Ahmed	Karachi
☐	Edit	Copy	Delete	Bilal Hussain	Islamabad
☐	Edit	Copy	Delete	Usman Tariq	Karachi
☐	Edit	Copy	Delete	Daniyal Raza	Islamabad
☐	Edit	Copy	Delete	Hassan Ali	Karachi

↑ ☐ Check all | With selected: Edit Copy Delete Export

Task A4 query showing employees from Karachi or Islamabad.

➤ Task A5

- This query displays female employees who are not part of the Engineering department.

SELECT *

FROM Employee

WHERE Gender = 'Female'

AND Department <>

'Engineering';

Filters Comparison & Logical Operators

Showing rows 0 - 2 (3 total, Query took 0.0006 seconds.)

```
-- Task A5 SELECT * FROM Employee WHERE Gender = 'Female' AND Department <> 'Engineering';
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate		
☐	Edit	Copy	Delete	2	Sara Ahmed	Female	Marketing	Marketing Officer	72000.00	Karachi	2020-07-10
☐	Edit	Copy	Delete	4	Maham Noor	Female	HR	HR Manager	88000.00	Lahore	2022-01-20
☐	Edit	Copy	Delete	8	Minal Fatima	Female	Sales	Sales Manager	99000.00	Lahore	2020-12-01

☐ Check all | With selected: Edit Copy Delete Export

Task A5 query showing female employees outside Engineering department.

➤ Task A6

- This query finds male employees earning between 70,000 and 90,000.

SELECT *

FROM Employee

WHERE Gender = 'Male'

AND Salary >= 70000

AND Salary <= 90000;

Showing rows 0 - 1 (2 total, Query took 0.0005 seconds.)

```
-- Task A6 SELECT * FROM Employee WHERE Gender = 'Male' AND Salary >= 70000 AND Salary <= 90000;
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate		
☐	Edit	Copy	Delete	7	Daniyal Raza	Male	Finance	Accountant	76000.00	Islamabad	2021-06-25
☐	Edit	Copy	Delete	9	Hassan Ali	Male	Marketing	SEO Specialist	81000.00	Karachi	2022-08-30

☐ Check all | With selected: Edit Copy Delete Export

Task A6 query showing male employees earning between 70,000 and 90,000.

➤ **Task A7**

- This query lists employees who are Software Engineers or have salary greater than 100,000.

SELECT *

FROM Employee

WHERE JobTitle = 'Software

Engineer'

OR Salary > 100000;

Showing rows 0 - 2 (3 total, Query took 0.0006 seconds.)

-- Task A7 **SELECT *** FROM Employee WHERE JobTitle = 'Software Engineer' OR Salary > 100000;

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

				EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate
☐	Edit	Copy	Delete	1	Ali Khan	Male	Engineering	Software Engineer	95000.00	Lahore	2021-03-15
☐	Edit	Copy	Delete	5	Usman Tariq	Male	Engineering	Senior Software Engineer	120000.00	Karachi	2018-09-12
☐	Edit	Copy	Delete	10	Maryam Imran	Female	Engineering	DevOps Engineer	105000.00	Lahore	2019-05-14

↑ ☐ Check all | With selected: Edit Copy Delete Export

Task A7 query showing Software Engineers or employees earning above 100,000.

➤ **Task A8**

- This query excludes employees from Marketing and Sales department.

SELECT *

FROM Employee

WHERE Department <>

'Marketing'

AND Department <> 'Sales';

Showing rows 0 - 5 (6 total, Query took 0.0005 seconds.)

```
-- Task A8 SELECT * FROM Employee WHERE Department <> 'Marketing' AND Department <> 'Sales';
```

[Edit inline](#) [\[Edit\]](#) [\[Create PHP code\]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

			EmpID	EmpName	Gender	Department	JobTitle	Salary	City	HireDate	
☐	Edit	Copy	Delete	1	Ali Khan	Male	Engineering	Software Engineer	95000.00	Lahore	2021-03-15
☐	Edit	Copy	Delete	4	Maham Noor	Female	HR	HR Manager	88000.00	Lahore	2022-01-20
☐	Edit	Copy	Delete	5	Usman Tariq	Male	Engineering	Senior Software Engineer	120000.00	Karachi	2018-09-12
☐	Edit	Copy	Delete	6	Areeba Khan	Female	Engineering	QA Engineer	85000.00	NULL	2023-04-18
☐	Edit	Copy	Delete	7	Daniyal Raza	Male	Finance	Accountant	76000.00	Islamabad	2021-06-25
☐	Edit	Copy	Delete	10	Maryam Imran	Female	Engineering	DevOps Engineer	105000.00	Lahore	2019-05-14

☐ Check all | With selected: Edit Copy Delete Export

Task A8 query excluding Marketing and Sales employees.

6. Conclusion

In this lab, different SQL filtering and sorting operations were performed successfully using MySQL in XAMPP. Various SQL clauses such as WHERE, AND, OR, BETWEEN, IN, LIKE, IS NULL, ORDER BY, and LIMIT were used to retrieve and manage employee records from the Employee table. This lab improved understanding of SQL query writing, data filtering techniques, pattern matching, and result sorting in.